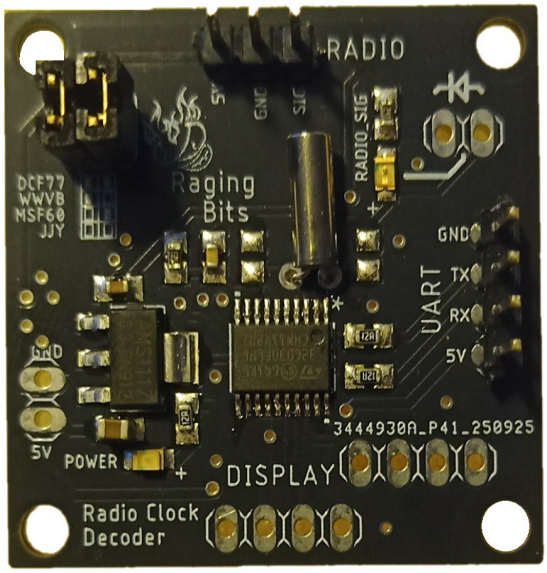
Raging Bits Atomic Clock Receiver module v1.0

Top level specs

Serial port interface.
With(out) screen.
4 Protocols
Robust fast time lock.



Device interface

Input Power - 5V Supply.
Serial interface 115200.
Screen – I2C SSD1608 128x64

Normal Work
UART Header – Communications for control and interface
TX – UART output data
RX – UART input data (Unused)
5V – 5 volts input power
GND – Ground

Serial Data format

The UART is fixed at 115200, 1 start bit, 1 stop bit, no control flow.
The data output happens on the exact second, using the following format.

Hour : minute : second[space] day / month / year [space]protocol
[2char][1char][2char][1char][2char][1char][2char][1char][2char][1char][4char][1char][4char]

Ex:

10:52:12 22/05/2028 WWVB
10:52:13 22/05/2028 WWVB
10:52:14 22/05/2028 WWVB
...

References

STM32G030Cx datasheet.
STM32G030Cx manual.
DCF77, WWVB, JJY, MSF60 - Wikipedia.